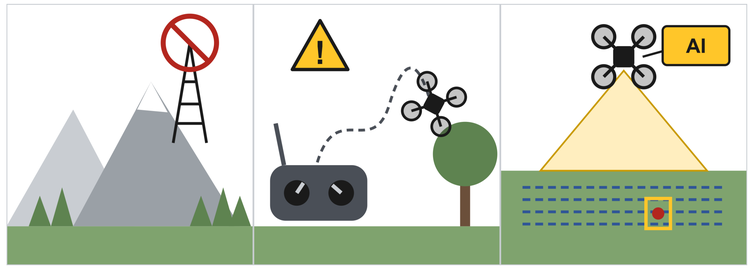
XX-000

KESTREL: Just Say Who's Missing

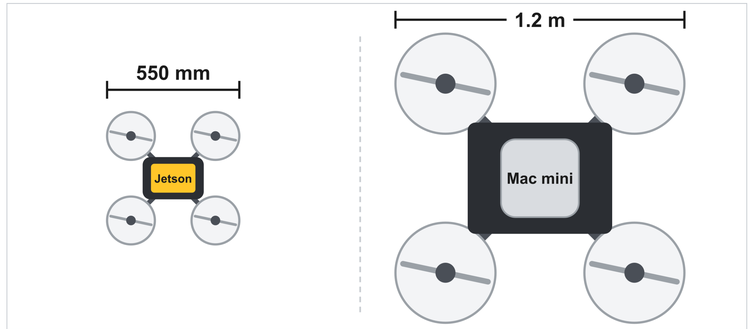
Shiva Shrestha and Honghui Xu

The Problem



No signal | Hard to fly | KESTREL flies itself

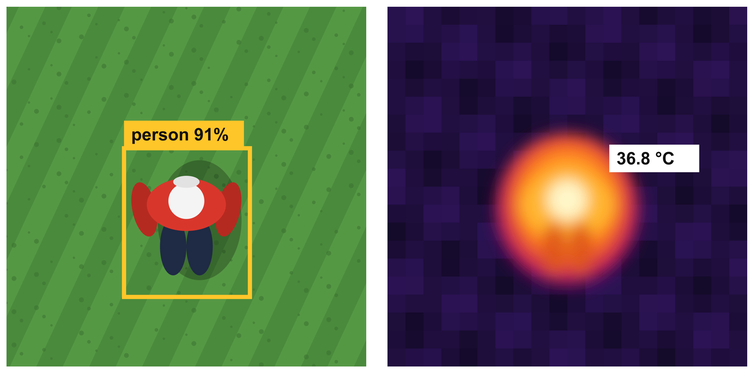
Two Drone Models



Compact, 550 mm
Jetson · Gemma 4 E2B

Large, 1.2 m
Mac mini · Qwen3.6-35B-A3B

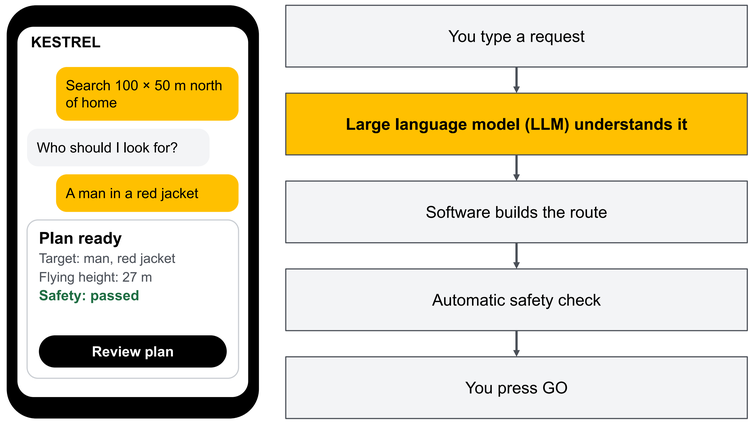
Finding the Person



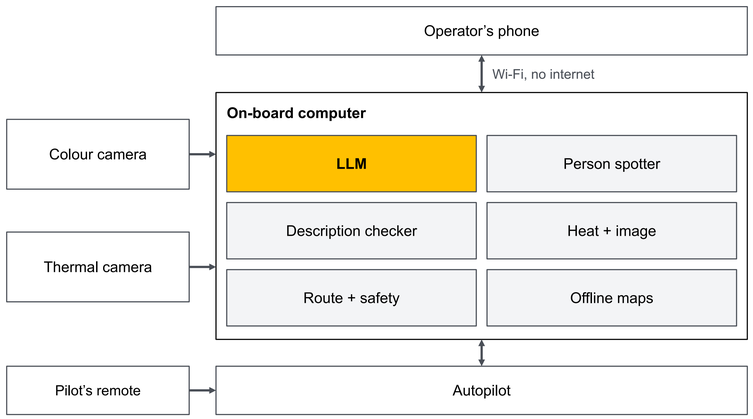
Colour camera | Thermal camera

MATCH → photo + map pin to phone

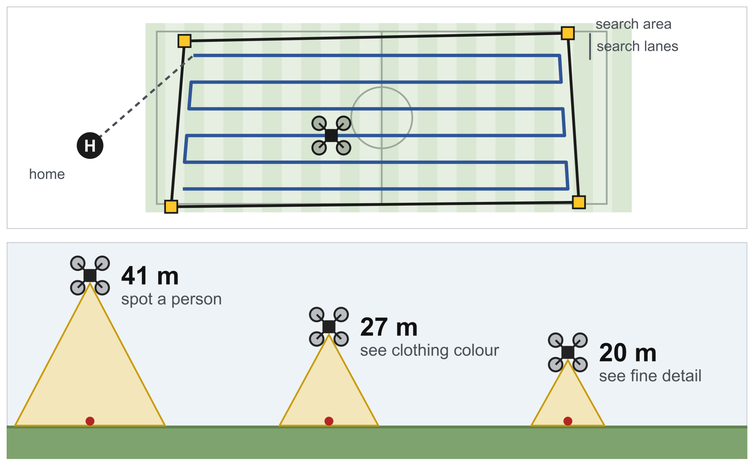
Plan a Search by Chatting



System Overview



Search Pattern and Height

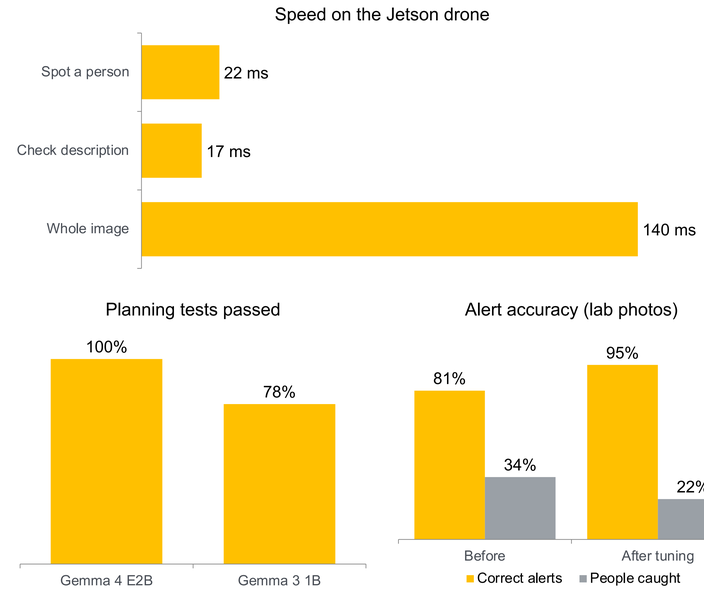


Automatic Safety Checks

- ✓ GPS locked
- ✓ Virtual fence
- ✓ Airspace approval
- ✓ Autopilot self-test
- ✓ Cameras and AI
- ✓ Battery level
- ✓ 400 ft limit
- ✓ Pilot's remote
- ✓ Route verified
- ✓ Legal checklist

FAA Part 107 · licensed pilot can take over

Results



631
software tests

0
internet needed

2
drone models

Shiva Shrestha
sshres16@students.kennesaw.edu
Honghui Xu (Advisor)
hxu10@kennesaw.edu

KENNESAW STATE UNIVERSITY
COLLEGE OF COMPUTING AND SOFTWARE ENGINEERING